Resource–Computation Separation in Controlled Realizations

Remarks on Information Debt and Substrate Skepticism

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Abstract

A previous resource-theoretic refinement of substrate skepticism introduced two exact stochastic information costs for finite controlled worlds: the on-path cost ρ_n, defined as the worst-policy Shannon entropy of the realized transcript, and the counterfactual common-seed cost χ_n, defined as the minimum entropy of one policy-independent exogenous seed whose deterministic response tree reproduces the complete policy-indexed interface. The present note asks what these quantities do *not* measure.

The central observation is a sharp separation. Every deterministic controlled world has $\rho_n = \chi_n = 0$ at every finite horizon, yet a succinctly represented deterministic world can have an NP-complete bounded goal-extension problem. Consequently, computational navigation can remain NP-hard on the exact zero-debt slice, and no tractability parameter that depends only on ρ_n, χ_n and their counterfactual premium can control the general extension problem unless P = NP. At unbounded horizon the separation is stronger still: deterministic systems with identically zero debt profiles may exhibit PSPACE-complete reachability, and Turing-complete deterministic dynamics yield undecidable eventual-reachability questions.

These are not new complexity phenomena in isolation. Hard deterministic planning, reversible PSPACE-complete dynamics, and the distinction between entropy and structural complexity have substantial prior literatures. The narrow contribution here is to locate those phenomena relative to the specific resource coordinates ρ_n and χ_n, and thereby expose a structural limitation of information-debt accounting: entropy over possible response structures does not measure the computational organization within a fixed response structure. Resource requirement is an invariant of the policy-indexed interface; computational accessibility additionally depends on how that interface is represented and queried.

1 Motivation and scope

The resource framework developed in [1] asks whether internally accessible stochastic structure can constrain realizations without identifying their substrate. For a finite controlled world W and horizon n, it defines

$$\rho_n(W) := \sup_{\pi \in \Pi_n} H(T_n^\pi), \tag{1}$$

$$\chi_n(W) := \min_{q \in \mathcal{Q}_n(W)} H_q(R_n), \tag{2}$$

where T_n^π is the transcript generated under a deterministic intervention policy π and R_n is a depth-n response tree. The feasible set $\mathcal{Q}_n(W)$ contains precisely those response-tree distributions whose deterministic policy readouts reproduce the target transcript law under every admissible policy. The earlier paper proves

$$\rho_n(W) \le \chi_n(W),$$

interprets χ_n operationally as a minimum policy-independent exogenous seed entropy, and shows that these costs are invariants of the internally accessible policy-indexed interface [1].

The present note concerns a different question:

> If a realization requires little or no stochastic information resource, does that make its counterfactual structure computationally easy to navigate?

The answer is no in a strong sense.

This distinction is adjacent to several established literatures and the scope should therefore be stated carefully. Computational mechanics has long distinguished entropy-like randomness from structural complexity [3, 4]; its input–output extension, the ϵ-transducer, gives minimal predictive models of stochastic transformations [5]. Classical deterministic planning is PSPACE-complete in general [6], and fully deterministic reversible systems can themselves have PSPACE-complete reachability problems [7]. Knowledge compilation explicitly separates representation succinctness from the tractability of queries over a compiled representation [8]. None of these facts is claimed as new here.

The narrower point is framework-specific: the exact resource coordinates (ρ_n, χ_n) may collapse to $(0, 0)$ while computational difficulty ranges from triviality to NP-completeness and, under broader unbounded questions, to PSPACE-completeness or undecidability. Thus the resource theory and the computational theory describe genuinely different axes.

2 Deterministic collapse of stochastic information debt

We use the same finite controlled-process setting as [1]. Fix finite action and observation alphabets $\mathcal{A}$ and $\mathcal{O}$. A world W is specified through horizon n by causal kernels

$$P_W(o_t \mid h_{t-1}, a_t), \qquad 1 \le t \le n.$$

Definition 1 (Deterministic controlled world)**.** *W is* deterministic through horizon n *if for every reachable history–action pair (h_{t-1}, a_t) there is an observation $o^\star(h_{t-1}, a_t)$ such that*

$$P_W(o^\star(h_{t-1}, a_t) \mid h_{t-1}, a_t) = 1.$$

Proposition 1 (Deterministic collapse)**.** *If W is deterministic through horizon n, then*

$$\boxed{\rho_n(W) = \chi_n(W) = 0.} \tag{3}$$

Consequently the counterfactual premium

$$\Delta_n^{\mathrm{cf}}(W) := \chi_n(W) - \rho_n(W)$$

also vanishes.

Proof. Fix a deterministic policy π. Because both the policy and the world are deterministic, the complete transcript T_n^π is a point mass. Hence $H(T_n^\pi) = 0$ for every π, and therefore $\rho_n(W) = 0$.

For the common-seed cost, determinism assigns a unique observation to every history–action node. Collecting these assignments produces a single response tree $r^\star \in \mathfrak{R}_n$ that reproduces the interface under every policy. Thus the point mass $q = \delta_{r^\star}$ belongs to $\mathcal{Q}_n(W)$ and has entropy zero. Since Shannon entropy is nonnegative, $\chi_n(W) = 0$. □

Proposition 1 is elementary but exposes the relevant blind spot. χ_n measures uncertainty over *which* response tree is selected by the exogenous source. It does not measure the description length, circuit complexity, search complexity, or dynamical depth of the unique tree when only one tree occurs.

In particular,

$$\boxed{\text{entropy over response structures} \neq \text{computational complexity within a fixed response structure.}} \tag{4}$$

3 NP-complete goal extension at exactly zero debt

We now make the separation explicit.

Definition 2 (Succinct Boolean controlled world)**.** *Given a Boolean formula $\varphi(x_1, \ldots, x_n)$, define the controlled world W_φ of horizon n as follows:*

- *the action alphabet is $\mathcal{A} = \{0, 1\}$;*
- *for $t < n$, the observation is deterministically $O_t = 0$;*
- *at the final time,*

$$O_n = \varphi(A_1, \ldots, A_n).$$

The description of W_φ contains φ succinctly rather than an explicit table of all 2^n action sequences.

The intermediate observations reveal no information and therefore every deterministic sequence

$$a = (a_1, \ldots, a_n) \in \{0,1\}^n$$

is implemented by an admissible deterministic policy.

Definition 3 (Goal-Extension)**.** *For the restricted family above,* Goal-Extension *asks whether there exists an admissible deterministic policy π such that the final observation is*

$$O_n = 1.$$

Theorem 1 (Zero-debt extension hardness)**.** Goal-Extension *is NP-complete even when restricted to controlled worlds satisfying*

$$\boxed{\rho_n(W) = \chi_n(W) = \Delta_n^{\mathrm{cf}}(W) = 0.} \tag{5}$$

Proof. Membership in NP is immediate for the restricted family: a witness is the n-bit action sequence $a = (a_1, \ldots, a_n)$, and evaluating $\varphi(a)$ is polynomial in the size of the formula.

For NP-hardness, reduce from 3SAT. Given a 3-CNF formula $\varphi(x_1, \ldots, x_n)$, construct W_φ as above. The construction is polynomial in $|\varphi|$. Then

$$\begin{aligned}
\textsc{Goal-Extension}(W_\varphi) = 1 &\iff \exists (a_1, \ldots, a_n) \in \{0,1\}^n : O_n = 1 \\
&\iff \exists (a_1, \ldots, a_n) : \varphi(a_1, \ldots, a_n) = 1 \\
&\iff \varphi \in \text{3SAT}.
\end{aligned}$$

The world W_φ is deterministic, so Proposition 1 gives $\rho_n(W_\varphi) = \chi_n(W_\varphi) = 0$, and hence $\Delta_n^{\mathrm{cf}}(W_\varphi) = 0$. □

The theorem should not be read as a new proof that deterministic planning or Boolean reachability can be hard. The reduction is deliberately elementary. Its role is to place NP-complete computational navigation on the exact zero-debt slice of the stochastic resource theory.

3.1 Why the hardness is not "hidden randomness"

The hardness in Theorem 1 is entirely endogenous. No stochastic seed is needed to realize W_φ. For each fixed policy the trajectory is fully determined, and one deterministic response tree fixes all counterfactual branches simultaneously.

The difficulty lies elsewhere: the succinctly represented deterministic map

$$(a_1, \ldots, a_n) \longmapsto \varphi(a_1, \ldots, a_n)$$

may encode a hard search problem. Randomness provenance and search complexity therefore need not track one another.

4 No tractability criterion from debt coordinates alone

The zero-debt slice yields an immediate parameterized consequence.

Because ρ_n, χ_n and Δ_n^{cf} are real-valued, define integer parameters

$$k_\rho(W) := \lceil \rho_n(W) \rceil, \qquad k_\chi(W) := \lceil \chi_n(W) \rceil, \qquad k_\Delta(W) := \lceil \Delta_n^{\mathrm{cf}}(W) \rceil.$$

Corollary 1 (Slice hardness)**.** Goal-Extension *remains NP-hard on the fixed slice*

$$k_\rho = k_\chi = k_\Delta = 0.$$

Consequently, unless $\mathrm{P} = \mathrm{NP}$*, none of these parameters alone can make the general problem fixed-parameter tractable.*

Proof. The reduction in Theorem 1 produces only instances on the displayed slice. An algorithm running in

$$f(k)\, |\langle W \rangle|^{O(1)}$$

for any one of the three parameters would therefore run in polynomial time on all reduced 3SAT instances, because $f(0)$ is a constant. □

The same argument gives a more general statement.

Corollary 2 (No debt-only tractability parameter)**.** *Let*

$$k(W) = g\big(\rho_n(W), \chi_n(W), \Delta_n^{\mathrm{cf}}(W)\big)$$

be any integer-valued parameter for which $g(0,0,0) = k_0 < \infty$. If the class of admissible instances contains the worlds W_φ, then Goal-Extension *cannot be fixed-parameter tractable with respect to k unless* P = NP.

This does not say that information debt is irrelevant to algorithms on restricted classes. It says only that the three debt coordinates, by themselves, do not control general computational accessibility. Additional structural parameters—treewidth, circuit class, causal width, compilation language, or other properties of the deterministic map—may do so.

5 Beyond NP: the zero-debt slice is computationally large

The bounded construction already establishes the main separation. Two standard complexity phenomena show that the gap becomes wider when the query or model class is enlarged.

5.1 PSPACE-complete deterministic reachability

General propositional STRIPS plan existence is PSPACE-complete [6]. More strikingly for the present purpose, fully deterministic and reversible systems can have PSPACE-complete reachability problems [7]. Such systems can be viewed as controlled worlds with deterministic transition kernels; in zero-player variants the action alphabet may even be a singleton.

For every finite horizon their stochastic information debt therefore obeys

$$\rho_n = \chi_n = 0.$$

The established PSPACE-completeness is not a contribution of this note. Its relevance is diagnostic: even reversibility and complete absence of exogenous stochasticity do not furnish a useful general upper bound on reachability complexity.

5.2 Undecidable eventual reachability

The separation extends past ordinary complexity classes.

Proposition 2 (Complete debt-profile blindness)**.** *There is a computably described family of deterministic singleton-action worlds for which*

$$\rho_n(W) = \chi_n(W) = 0 \qquad \textit{for every finite } n, \tag{6}$$

while the eventual-reachability question

$$\exists t \geq 0 :\ O_t = 1?$$

is undecidable.

Proof. Given a deterministic Turing machine M and input x, construct a singleton-action world whose internal state is the configuration of $M(x)$. Each world step applies one transition of M. Set $O_t = 1$ exactly when the current configuration is halting, and $O_t = 0$ otherwise.

The world is deterministic, so Proposition 1 gives Eq. (6) for every finite horizon. Eventual reachability of $O_t = 1$ is equivalent to asking whether M halts on x, which is undecidable [10]. □

Thus even the entire debt profile

$$n \longmapsto (\rho_n(W), \chi_n(W))$$

may be identically zero while an unbounded dynamical question is undecidable.

This statement concerns Shannon seed entropy and should not be conflated with topological entropy. As a useful comparison, recent work also exhibits universal Turing machines with zero topological entropy [9]; that is a distinct entropy notion and is cited only as related evidence for the broader independence of dynamical computation from simple entropy diagnostics.

6 Resource magnitude, resource evaluation, and navigation

The combined picture contains at least three different computational or resource questions.

(i) **Resource magnitude:**

$$R_n(W) := \chi_n(W).$$

How much policy-independent exogenous entropy is required by an exact deterministic common-seed realization?

(ii) **Resource-evaluation complexity:** how difficult is it, from a succinct description of W, to compute or optimize $\chi_n(W)$ itself? Minimum-entropy coupling, which appears as a special case of exogenous-entropy minimization, is NP-hard [2].

(iii) **Navigation complexity:** how difficult is it to find or decide the existence of a policy, history, preimage, or trajectory satisfying a target condition? Theorem 1 shows that this can be NP-complete even when the resource magnitude is exactly zero.

These should not be collapsed into one notion of "complexity." A quantity may be easy to define and hard to optimize; a realization may require no stochastic resource and still encode hard search; a high-entropy source may, once supplied, be computationally trivial to relay.

For example, n independent fair output bits with no interventions have

$$\rho_n = \chi_n = n$$

in the framework of [1], while merely generating the outputs after an n-bit random source has been supplied requires no difficult search. High debt therefore does not itself imply computational hardness, just as zero debt does not imply computational ease.

The safest general conclusion is not a numerical "orthogonality" statement, but an incomparability of resource requirement and computational navigation without further structural assumptions.

7 Presentation relativity of computational accessibility

There is a second asymmetry between the resource and computational sides.

The previous paper proves that if two realizations have the same complete policy-indexed internal interface, then

$$\rho_n(R_1) = \rho_n(R_2), \qquad \chi_n(R_1) = \chi_n(R_2).$$

The costs are therefore semantic interface invariants.

Computational complexity, by contrast, is defined for encoded problem instances. Let $\mathcal{E}$ be a representation scheme for controlled worlds and let $\mathcal{Q}$ be a class of allowed queries. Computational accessibility should therefore be written schematically as

$$C_{\mathrm{acc}}(W; \mathcal{E}, \mathcal{Q}),$$

rather than as a presentation-free number $C_{\mathrm{acc}}(W)$.

The familiar Boolean example makes the point. A Boolean function may be presented succinctly by a CNF formula or explicitly by its truth table. The underlying function is the same, but scanning an explicit truth table is polynomial in the table size, while satisfiability for succinct CNF input is NP-complete. The exponential cost has moved into the representation size.

This is precisely the distinction emphasized in knowledge compilation: compiled representations are compared both by their succinctness and by which queries and transformations they support in polynomial time [8]. A claim that a compatible realization is "computationally accessible" is therefore incomplete until the representation and query model are fixed.

For substrate skepticism the contrast is conceptually useful:

$$\boxed{\begin{array}{ll} \text{stochastic resource requirement:} & \text{interface-invariant,} \\ \text{computational accessibility:} & \text{presentation- and query-relative.} \end{array}} \tag{7}$$

8 Relation to structural complexity

The conclusion that entropy and computation are different axes has substantial precedent. Crutchfield and Young introduced statistical complexity explicitly as a quantity distinct from information-theoretic entropy [3]. Computational mechanics later characterized causal states as minimal sufficient predictive states [4], and the ϵ-transducer extends this program to stochastic input–output processes [5].

Accordingly, no novelty is claimed here for the general idea of separating randomness from structure, nor for future-equivalence constructions of predictive states. The present note instead uses the exact resource measures ρ_n and χ_n for a different operational question: how much exogenous stochastic information is required to realize a complete controlled interface?

The zero-debt theorem shows that these resource monotones deliberately ignore a dimension that structural-complexity theories attempt to capture. A deterministic response tree has zero seed entropy even if its internal counterfactual organization is computationally difficult to navigate.

This is not a defect of ρ_n or χ_n. It is a boundary of their meaning.

9 Implications for substrate skepticism

The earlier resource paper sharpened substrate skepticism by showing that internal stochastic structure can constrain realization resources without identifying the actual substrate [1]. The present separation adds a third distinction.

For a realization map or controlled presentation one may ask:

(a) *Existence:* does a compatible realization or goal-achieving history exist?

(b) *Identification:* which compatible realization is the actual substrate?

(c) *Accessibility:* can an embedded or external reasoner construct, navigate, or recover a compatible realization efficiently from the available representation?

These questions need not have the same answer.

Substrate skepticism concerns primarily the second: internally equivalent realizations need not reveal which realization is actual. Information-debt bounds constrain the first at the level of resource feasibility: some realization classes may be excluded because they cannot supply sufficient exogenous entropy. Theorem 1 concerns the third and shows that even when the resource obstruction disappears completely, computational accessibility need not follow.

Thus:

$$\boxed{\text{realization feasibility} \not\Rightarrow \text{realization identification} \not\Rightarrow \text{efficient computational navigation},} \tag{8}$$

where the arrows are schematic non-implications rather than a claim that the three notions form a strict logical chain in every formalization.

This also clarifies why the resource framework does not decide P versus NP. NP-hardness is already present in the deterministic zero-debt sector. A hypothetical additional principle that turned every succinct NP-universal compatible-realization problem into a polynomial-time navigable representation would indeed have major complexity consequences, potentially including P = NP. Nothing in the Shannon resource bounds supplies such a principle.

10 Novelty status and claim boundary

Because this note sits at the intersection of several mature literatures, its claim boundary should be explicit.

(i) It is *not* claimed as new that deterministic planning or reachability can be NP-hard or PSPACE-complete [6, 7].

(ii) It is *not* claimed as new that entropy and structural complexity are distinct notions [3, 4].

(iii) It is *not* claimed as new that representation succinctness and query tractability must be distinguished [8].

(iv) The narrower contribution is the explicit separation relative to the exact stochastic realization coordinates introduced in [1]:

$$\rho_n = \chi_n = 0 \quad \text{is compatible with NP-complete bounded extension,}$$

together with the immediate fixed-slice consequences for any debt-only tractability parameter and the distinction between interface-invariant resource requirements and presentation-relative computational accessibility.

A literature search can never prove absolute priority. The defensible claim is therefore framework-relative rather than historical: the results above are derived explicitly for the ρ_n/χ_n information-debt formalism, while the broader complexity phenomena on which they rest are acknowledged as prior art.

11 Conclusion

The resource theory of exact stochastic realization and the computational theory of navigating a realization answer different questions.

For deterministic controlled worlds,

$$\rho_n = \chi_n = 0,$$

because no exogenous randomness is needed to choose the realized transcript or the complete response tree. Nevertheless, a succinct deterministic response structure can encode NP-complete bounded extension, PSPACE-complete reachability, or—under unbounded Turing-complete dynamics—undecidable eventual reachability.

The central separation can therefore be stated concisely:

$$\boxed{\begin{array}{c}\text{Information debt measures how much stochastic resource a realization must supply. It}\\ \text{does not measure how difficult a deterministic realization is to navigate, analyze, or invert.}\end{array}} \tag{9}$$

This observation complements rather than revises the original substrate argument. Exact stochasticity constrains the provenance of information. Computational accessibility requires additional structural and representational information that the stochastic resource coordinates were never designed to encode.

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